GRABS: Graded Satisfaction and Belief-Guided Search for Demand-Driven Navigation

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Abstract*— Demand-driven navigation (DDN) asks a robot to satisfy an implicit request such as "I am thirsty" with any suitable object, rather than the named target of object goal navigation. Yet water, juice, and coffee satisfy it unequally, so the first acceptable object may not be the best. We formulate *best-available demand-driven navigation*: retrieving the highest-satisfaction object present while balancing quality against time, generalizing both. We present GRABS, a belief-guided framework that needs only a room-level floor plan. A large language model grounds the request into candidate categories with graded satisfaction probabilities, and a second query estimates each candidate's likelihood per room; these define a room utility, the expected satisfaction from searching each room, that a time-aware Markov decision process (MDP) balances against travel and search time. A vision-language model (VLM) verifies candidates, and an unsuccessful search lowers that room's belief before replanning. Holding an acceptable but suboptimal object, the robot asks whether to continue only when finding a better one would exceed a patience threshold. In AI Habitat, Isaac Sim, and hardware trials, GRABS matches the best baselines on explicit requests, raises absolute best-object accuracy by 27.8%, 29.6%, and 27.1% on HM3D, MP3D, and Isaac Sim over a first-acceptable-object baseline, and its time-aware MDP reaches them in up to 29.9% less time than heuristic planning. On hardware, it retrieves the best object for every vague request and asks the user when required.

I. Introduction

Demand-driven navigation (DDN) [1] asks a robot to satisfy an implicit request such as "I am thirsty," succeeding when it reaches any object that meets the need. It relaxes object goal navigation [2], where the user names the target category and the robot only has to locate it, together with its open-vocabulary and language-guided variants [3], [4] that accept richer descriptions. Because people state a purpose rather than an object [5], several objects may satisfy the request, such as water, tea, or coffee, and rarely to the same degree. This turns navigation into a decision problem (Figure 1): stopping at the first acceptable object can return a poor result, whereas searching every room imposes a delay only the user can judge acceptable. The robot must therefore infer which objects would serve the purpose, judge where they are likely to be, and weigh whether further search is worth the time. Water may be preferable to coffee, yet coffee in the next room can outweigh water several rooms away once the user's patience is considered.

DDN extensions cover multi-object demands, VLM-based reasoning, and hidden-target inference [6]–[8]. All of them, however, treat satisfaction as *binary*: an object either satisfies the request or does not. A cup of coffee and a bottle of water thus count as equally good answers to "I am thirsty." None chooses among substitutable candidates that satisfy the request to different degrees when their availability and locations are unknown, or weighs a better candidate against the time needed to reach it.

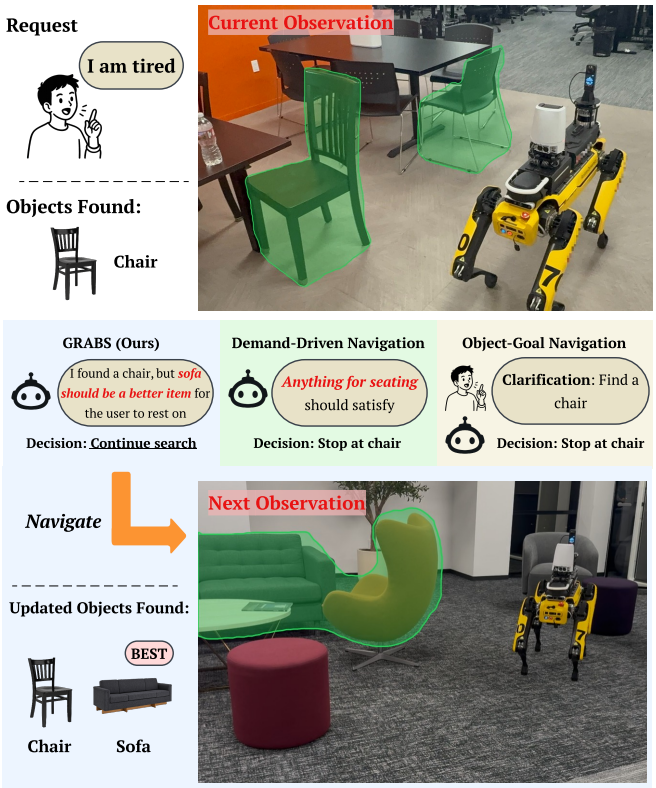

Fig. 1: Given "I am tired," conventional object goal navigation searches for a clarified target, while demand-driven navigation stops after finding any satisfactory object. GRABS recognizes that the chair is acceptable but continues its belief-guided search for the higher-satisfaction sofa.

We formulate this problem as *best-available demand-driven navigation*. It replaces DDN's binary success criterion with a graded satisfaction value for each candidate object, so the robot must retrieve the highest-satisfaction object actually present within an acceptable time. We consider standard DDN and explicit object goal navigation as special cases with binary satisfaction.

We introduce GRABS (GRAded satisfaction and Belief-guided Search), a framework for this task, summarized in Figure 2. It requires only a room-level floor plan with semantic labels, commonly available for buildings [9], and

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no task-specific training. GRABS uses an LLM's commonsense capability for two judgments: one call grounds the request into candidate categories with graded satisfaction probabilities and a visual verification criterion, and a second produces the robot's initial belief over which rooms contain each candidate. Satisfaction probabilities and beliefs combine into a utility for searching each room, and a time-aware Markov decision process (MDP) selects the next room with lookahead over the remaining search. After each observation, a VLM detects and verifies candidates, and Bayesian inference down-weights rooms searched without success before the robot replans.

When the robot holds an acceptable but suboptimal object, it estimates from the updated belief the additional time required to find a better one. It continues automatically when this time is small and asks the user whether to continue when it exceeds a patience threshold derived from a user survey. The user may instead tell the robot which object to look for. In practice, however, users in an unfamiliar environment often do not know which objects are present [1] and therefore cannot name the best available one. When the user can, the request is explicit object goal navigation, which already falls within our framework.

To demonstrate the efficacy of GRABS, we evaluate it at three levels of increasing realism: with oracle perception in AI Habitat [10] on the HM3D [11] and MP3D [12] datasets, with VLM perception in an Isaac Sim [13] digital twin of a physical office, and in hardware trials on a quadruped robot [14] in that office. Across explicit and vague-intent requests, we compare against five baseline planners and report success rate, path efficiency, best-object retrieval, and search time. The hardware trials further exercise graded candidate selection, attribute and relation verification, belief-guided replanning, and time-aware interaction with the user.

Our main contributions are:

- We formulate best-available demand-driven navigation, which extends DDN to graded satisfaction so that the robot seeks the best object present rather than any acceptable one, with DDN and explicit object goal navigation as special cases.
- We introduce a grounding representation that separates request satisfaction from room occurrence and combines them into a room utility, which lets the planner compare rooms holding different candidates and revise presence beliefs without altering satisfaction.
- We develop a time-aware MDP with lookahead room selection, Bayesian belief updates after misses, and replanning after each observation, which decides whether searching for a better object is worth the time.
- We evaluate GRABS at three levels of increasing realism, from oracle-perception simulation through a digital twin with VLM perception to hardware trials.

II. Related Work

A. Object Goal Navigation

Object goal navigation [2] requires an agent to find an instance of a named category in an unseen environment. Learning-based policies [15] and modular map-based planners [16], [17] were followed by zero-shot methods that use language and vision-language models as semantic priors over frontiers or scene graphs [18]–[22]. Open-vocabulary benchmarks and methods extend the target from a fixed label set to free-form language [3], [4]. These methods plan over frontiers of an online metric map, whereas planning over a room-level floor plan reduces the search to a small graph solvable exactly at every step. SEEK [9] plans over a scene graph with semantic object–room priors and probabilistic reasoning, but searches for a single predefined inspection target. In all of these, the language names the object, whereas GRABS must first infer candidates, rank their satisfaction, and select the best available one.

B. Demand-Driven Navigation

Demand-driven navigation [1] is closest to ours. The agent receives an implicit request such as "I am thirsty" and succeeds upon reaching any object that satisfies it. Follow-up work extends the task to multi-object demands with personal preferences [6], replaces learned policies with VLM reasoning [7], and grounds or navigates to a single hidden target implied by an intention [5], [8], while personalized object search conditions a specified target on a user's placement habits [23]. Language-guided navigation benchmarks specify a route, a destination, or a referred object [24]–[27], and foundation-model planners execute such instructions zero-shot by reasoning over observations, scene graphs, or landmark sequences [28]–[30].

In all of this work, satisfaction is binary, and no formulation asks whether an acceptable object in hand is the best available, weighs further search against time, or consults the user. The language-guided systems likewise reduce each instruction to one target or landmark plan. Their success metrics reward reaching any satisfying object and therefore cannot distinguish an acceptable object from the best one, which motivates a graded evaluation of the retrieved object. GRABS instead assigns graded satisfaction to substitutable categories and uses it throughout planning and termination.

C. Belief-Aware Search and Human Interaction

Probabilistic object search maintains a belief over target locations [31] and selects actions by detection likelihood [32], information gain [33], or travel cost [9], [34], typically formulated as an MDP or a partially observable MDP (POMDP). These methods fix the target category, so their belief models only where the target is likely to occur and never how well a found object satisfies the request. GRABS separates these questions. A satisfaction probability models how well each category addresses the request, and a spatial belief models where it occurs. Together they form a room utility, the expected satisfaction the robot obtains by searching that room, and Bayesian updates after misses revise where candidates are likely to be, not how well they satisfy the request.

Interactive navigation uses dialogue to disambiguate a referred instance among distractors [35], and uncertainty-

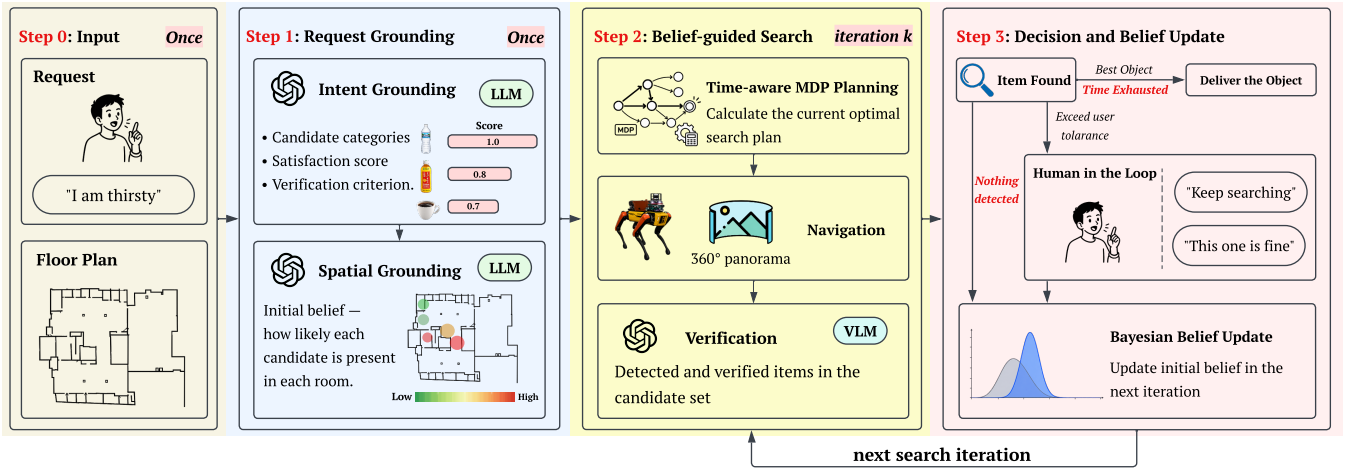

Fig. 2: Overview of GRABS. Given a vague user request and a room-level floor plan with semantic room labels, the framework grounds the user's intent and initializes a spatial belief over candidate objects. It then iteratively plans, navigates, observes, and verifies candidates. The decision-making step determines whether to deliver the current candidate, consult the user, or update the belief and continue searching.

aligned LLM planners ask for help when they cannot choose an action [36]. Stopping rules for embodied question answering end exploration once calibrated answer confidence reaches a threshold [33]. In GRABS, the user has no hidden target. The robot asks only whether the expected improvement from further search is worth the wait, so interaction controls when to stop rather than which object is correct.

III. Methodology

GRABS first grounds the request into candidate objects and spatial beliefs, then cycles through utility-based room planning, observation, decision making, and belief updates.

A. Problem Formulation

In demand-driven navigation, the user expresses a need rather than naming an object category. We extend it from binary to graded satisfaction. For example, "I am thirsty" admits several objects of differing suitability. The robot must infer these candidates and retrieve the best available one while accounting for search time and the user's willingness to wait.

At the start of an episode, the robot receives request r and a room-level floor plan $F = (V, E)$. Here, V contains semantically labeled room nodes and $E \subseteq V \times V$ encodes traversable connections. The floor plan does not provide object locations or states.

Let X be the set of physical object instances. The ground-truth request-satisfaction probability $S_r(x) \in [0,1]$ gives the probability that delivering $x \in X$ satisfies r. The ideal outcome is the available object with the highest satisfaction:

$$x^\star \in \arg\max_{x \in X} S_r(x). \tag{1}$$

Because X is unknown, the robot must infer and search for suitable object categories while balancing satisfaction and search time. It may also consult the user when continued search could improve the result but increase waiting. Standard DDN is the special case $S_r(x) \in \{0,1\}$ with several acceptable categories; explicit object goal requests restrict it further to a single category, where 1 denotes an exact category-and-state match.

B. Request Grounding

1) Intent Grounding: Intent grounding runs once before navigation. Given r and the semantic room labels in F, the LLM generates candidates that are both relevant to the request and plausible in the environment, without observing ground-truth room contents:

$$(G, \phi_r) = f_{\text{item}}(r, F), \qquad G = \{(c_i, u_i)\}_{i=1}^{M}. \tag{2}$$

Here f_{item} denotes the intent-grounding LLM call, and the set G contains M distinct object categories c_i. Each u_i is the LLM-estimated probability that an instance of c_i satisfies r, serving as a category-level estimate of $S_r(x)$. Categories the LLM scores at zero are omitted, so $u_i \in (0,1]$. The value u_i measures request satisfaction, not environmental occurrence. The shared criterion ϕ_r specifies the observable attributes, states, and relations required to verify the request.

2) Spatial Grounding: A second LLM call, f_{room}, maps G and F to the initial spatial belief

$$B_0 = f_{\text{room}}(G, F) = \{b_0(c_i, v) \mid v \in V\}_{i=1}^{M}, \tag{3}$$

where $b_0(c_i, v) \in [0,1]$ is the inferred probability that room v contains at least one instance of c_i. This inference uses semantic room labels but not ground-truth contents. Figure 3 shows simplified versions of both grounding prompts with example outputs for "I am very thirsty."

At iteration k, with $b_k(c_i, v)$ the current belief, candidates are ordered by decreasing u_i. We assume that the presence of one category in a room is independent of the presence of any other. Under this assumption, the expected request-satisfaction probability for room v, which serves as the room

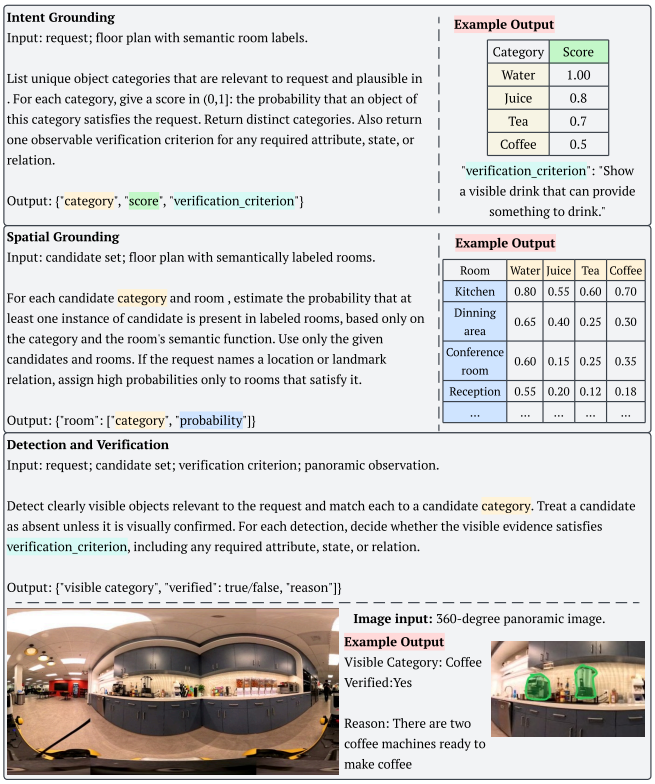

Fig. 3: Simplified prompts for intent grounding, spatial grounding, and VLM-based detection and verification, with example outputs for "I am very thirsty."

utility, is

$$U_k(v) = \sum_{i=1}^{M} u_i\, b_k(c_i, v) \prod_{j=1}^{i-1} \left(1 - b_k(c_j, v)\right). \tag{4}$$

The product is the probability that no higher-ranked candidate is present, so each term represents the probability that c_i is the best candidate in v, weighted by u_i. Thus, $U_k(v) \in [0,1]$. At the start of the search, $U_0(v)$ is computed from the initial beliefs B_0. Both grounding outputs are initial estimates that are updated after each observation during the search.

C. Belief-Guided Search

1) Time-Aware MDP Planning: At iteration k, the planner uses the room utility $U_k(v)$ as the probability of satisfying the request by searching room v. For efficient online replanning, the MDP assumes that any satisfying candidate present in a room is verified successfully. Perception misses are handled after execution by the Bayesian update.

We formulate room selection as an MDP (S, A, P_k, T). The state space $S = V \cup \{s_G\}$ contains the room states and an absorbing goal state $s_G \notin V$. A state $s \in V$ is the robot's current room, and s_k is its room at iteration k. Action a_v navigates to and searches a reachable room v, and $A(s)$ denotes the set of actions available in s. The transition model is $P_k(s' \mid s, a_v)$.

The MDP time cost is $T(s, a_v) = t_{\text{travel}}(s, v) + t_{\text{obs}}$, the time to travel from s to v along the floor plan at the robot's nominal speed v_{rob}, plus a fixed room-inspection time t_{obs}.

Searching v reaches s_G with probability $U_k(v)$ and otherwise leaves the robot in v:

$$\begin{aligned} P_k(s_G \mid s, a_v) &= U_k(v), \\ P_k(v \mid s, a_v) &= 1 - U_k(v). \end{aligned} \tag{5}$$

The expected discounted cost of a_v is

$$Q_k(s, a_v) = T(s, a_v) + \gamma(1 - U_k(v))\, J_k(v), \tag{6}$$

where $\gamma \in (0,1)$ is a discount factor that makes value iteration a contraction and guarantees convergence with revisits allowed, $J_k(v)$ is the remaining cost after failure, and $J_k(s_G) = 0$. The optimal cost-to-go is

$$J_k(s) = \min_{a_v \in A(s)} Q_k(s, a_v). \tag{7}$$

A smaller $J_k(s)$ indicates less expected discounted time to satisfy the request. Value iteration computes the minimizing policy $\pi_k(s)$. Beliefs are held fixed within each solve, since updating them inside the solve would make real-time execution infeasible. Only $\pi_k(s_k)$ is executed before the Bayesian update and replan. Room revisits are allowed.

2) Navigation, Observation, and Verification: The selected action $\pi_k(s_k) = a_{v_k}$ sends the robot to room v_k, where it captures an approximately 360° panoramic observation z_k. The VLM returns the verified set

$$C_k = f_{\text{detect}}(z_k, G, \phi_r) = \{c_i\}_{i=1}^{N_k}, \tag{8}$$

where f_{detect} denotes the VLM detection and verification call and N_k is the number of verified categories. Each $c_i \in C_k$ retains its u_i from G. The bottom panel of Figure 3 shows this prompt and an example verification.

A relevant category outside G may be scored by the VLM, verified against ϕ_r, and added to both C_k and G. The robot maintains g_k, the highest satisfaction probability found so far. After observation, g_{k+1} is the larger of g_k and the highest u_i in C_k. If no better candidate is verified, the current object is retained. An undetected candidate or one that fails ϕ_r is treated as a miss.

D. Decision Making and Belief Update

1) Decision Making: Let $u^\star = \max_i u_i$ be the highest satisfaction probability in G. If $g_{k+1} = u^\star$, the robot delivers the verified object. Otherwise, the MDP is re-solved with the updated beliefs over candidates that would improve on g_{k+1}, and $T_{k+1} = J_{k+1}(s_{k+1})$ is the expected time to find one.

The search ends at the mission budget $T_{\max}$, delivering the current object if one exists and otherwise reporting failure. If the robot holds a satisfactory but suboptimal object, it compares T_{k+1} with the user patience threshold T_h, obtained from a questionnaire. When $T_{k+1} > T_h$, it asks whether to continue or deliver the current object and follows the response subject to $T_{\max}$. Otherwise, the search continues.

TABLE I: Summary of the three experimental settings.

	AI Habitat	Isaac Sim	Hardware
Environment	36 HM3D + 11 MP3D	Simulated office	Physical office
Observation	Ground-truth inventory	360° panorama	360° observation
Detection method	Oracle	VLM	VLM
Tests per method	576 + 176	96	5
Human interaction	Disabled	Disabled	Enabled ($T_h = 56\,\mathrm{s}$)

TABLE II: Request types used in the evaluation.

Type	Expected target	Example
Explicit Requests		
Explicit	Named item.	"Find the laptop."
Attribute	Item with a visual attribute.	"Find the blue backpack."
Relational	Item related to an anchor.	"Find a silver trash bin near a door."
Verification	Item in a specified state.	"Is the TV on?"
Vague-Intent Requests		
Affordance	Affords a required function.	"I need something to watch a presentation with a group."
Human need	Addresses a human condition.	"I am very thirsty."
Situational	Satisfies a situational demand.	"I spilled some coffee on the table."
Preference	Best matches a preference.	"Which seat would be most comfortable for resting?"

2) Bayesian Belief Update: For subsequent planning, categories with $u_i \leq g_{k+1}$ are excluded and $U_{k+1}(v)$ is recomputed over the remaining categories. For each such c_i not verified in searched room v_k, the observation is treated as a possible miss. Assuming negligible verified false positives and perception recall η, the posterior belief after a missed detection is

$$b_{k+1}(c_i, v_k) = \frac{(1-\eta) b_k(c_i, v_k)}{1 - \eta b_k(c_i, v_k)}. \tag{9}$$

An undetected candidate or one that fails ϕ_r receives this update. Beliefs for unsearched rooms remain unchanged. Because a category may occur in multiple rooms, its per-room beliefs are treated as independent and are not normalized across rooms. Categories newly added to G during verification, as described in Section III-C.2, are included in this recomputation.

IV. Experimental Setup

We evaluate GRABS in AI Habitat, Isaac Sim, and a physical office, progressing from oracle perception to VLM perception and full hardware execution.

A. Request Types

We evaluate four explicit and four vague-intent request types, summarized in Table II. Each of the eight types is instantiated twice per scene in AI Habitat ($36 \times 8 \times 2 = 576$ and $11 \times 8 \times 2 = 176$ tests per method) and four times from each of three starting positions in Isaac Sim ($8 \times 3 \times 4 = 96$). The vague-intent types correspond to demand instructions in DDN. For AI Habitat, GPT-5.6-sol, with reasoning enabled, generates requests and reference answers offline, which we manually review and correct. Explicit requests have one target, while outputs from repeated vague-intent queries are combined into graded candidate sets. Isaac Sim and hardware requests are constructed directly from available objects. At runtime, all LLM and VLM calls use GPT-5.6-luna with reasoning disabled and temperature 0.

B. Environment Construction

Table I summarizes the three settings. AI Habitat isolates planning under oracle perception, Isaac Sim introduces VLM perception, and the hardware trials evaluate the complete system with human interaction.

AI Habitat contains 36 HM3D and 11 MP3D scenes. We annotate each room's semantic label and navigation location, then compute inter-room shortest paths using the Habitat PathFinder API. Each scene has a text inventory containing ground-truth objects and distractors generated by GPT-5.6-sol, which we manually review and correct. Object presence is sampled from room-occurrence probabilities assigned by the same model. Each sampled item is removed with probability 0.3 to create absent targets, which test belief updates and replanning. Rooms contain 21 items on average, and an oracle search reveals the complete inventory.

Isaac Sim uses a digital twin of the physical office, with manually placed objects and rendered 360° panoramas. Hardware experiments use a Boston Dynamics Spot quadruped [14] with a custom sensor payload. The physical office is left in its existing state. The hardware evaluation uses the four vague-intent examples and the relational explicit example listed in Table II. Across all settings, $v_{\mathrm{rob}} = 0.5\,\mathrm{m/s}$, $t_{\mathrm{obs}} = 30\,\mathrm{s}$, $T_{\max} = 600\,\mathrm{s}$, $\gamma = 0.95$, and $\eta = 0.8$.

C. Evaluation Protocol

For explicit requests, we report success rate (SR), success weighted by path length (SPL) [2], traveled distance, and completion time. For vague-intent requests, Best and Top-2 report how often the retrieved object is the highest-scoring candidate or among the top two. Mean satisfaction (MS) is the mean satisfaction probability of the retrieved object, and time is the mean search duration.

D. Comparison Methods

We compare GRABS with five methods:

- **Simple planner:** Our MDP ablation. It selects the room with the highest heuristic utility-to-distance ratio instead of solving the MDP, following the utility-over-cost frontier selection used in zero-shot object goal navigation [20], [34].
- **No-intent planner:** Scores rooms directly from the request without graded candidate objects. Most existing methods for implicit requests operate this way, stopping at the first object that satisfies the request [1], [7], [30].
- **Direct GPT planner:** Generates a fixed room order without spatial beliefs or replanning, as in LLM-generated navigation plans [37]–[39].
- **Landmark planner:** Grounds the request to one landmark, as in methods that resolve a free-form request to a single target [21], [30], [40].
- **Random planner:** Visits rooms in a seeded random order, prioritizing unvisited rooms, the room-level analogue of target-agnostic frontier exploration [4], [9].

TABLE III: AI Habitat results for explicit and vague-intent requests, reported as **HM3D / MP3D**. Distance is in meters, time in seconds, and MS is mean satisfaction. Starred variants use LLM-based continuation and apply to the vague-intent columns only.

Mode	Explicit Requests				Vague-Intent Requests			
	SR (%) ↑	SPL (%) ↑	Dist. (m) ↓	Time (s) ↓	Best (%) ↑	Top-2 (%) ↑	MS ↑	Time (s) ↓
Simple planner	96.9 / 97.7	80.3 / 85.0	15.4 / 21.0	52.4 / 57.5	73.6 / 69.3	78.1 / 85.2	0.786 / 0.840	140.3 / 122.7
No-intent planner	94.1 / 96.6	77.3 / 84.6	16.3 / 19.7	55.2 / 54.5	48.6 / 44.3	60.8 / 55.7	0.619 / 0.588	29.9 / 30.1
Direct GPT planner / ⋆	92.0 / 87.5	70.1 / 79.4	18.4 / 20.4	37.0 / 28.3	63.5 / 56.8	71.5 / 71.6	0.735 / 0.704	45.8 / 34.5
Landmark planner / ⋆	90.6 / 89.8	76.7 / 80.8	14.6 / 17.8	49.5 / 49.5	64.9 / 65.9	73.6 / 73.9	0.749 / 0.759	81.6 / 79.2
Random planner / ⋆	91.3 / 89.8	29.8 / 33.7	56.6 / 68.5	132.4 / 131.9	56.9 / 53.4	72.6 / 65.9	0.736 / 0.691	136.1 / 151.0
GRABS (Ours)	96.5 / 97.7	80.0 / 88.3	15.7 / 19.4	54.1 / 56.9	76.4 / 73.9	80.2 / 88.6	0.803 / 0.880	98.4 / 86.0

TABLE IV: Isaac Sim results for explicit and vague-intent requests. Distance is in meters, time in seconds, and MS is mean satisfaction. Starred variants use LLM-based continuation and apply to the vague-intent columns only.

Mode	Explicit Requests				Vague-Intent Requests			
	SR (%) ↑	SPL (%) ↑	Dist. (m) ↓	Time (s) ↓	Best (%) ↑	Top-2 (%) ↑	MS ↑	Time (s) ↓
Simple planner	89.6	85.5	40.6	76.2	64.6	81.2	0.799	166.0
No-intent planner	91.7	86.7	25.9	49.6	37.5	60.4	0.606	25.4
Direct GPT planner / ⋆	89.6	87.4	30.2	25.1	45.9	62.5	0.708	63.9
Landmark planner / ⋆	81.2	78.8	24.6	53.7	39.6	72.9	0.682	90.4
Random planner / ⋆	37.5	19.0	102.7	155.2	27.1	66.7	0.578	127.0
GRABS (Ours)	93.8	89.6	29.1	57.6	64.6	83.3	0.805	120.8

The Direct GPT, Landmark, and Random planners retain no graded candidates, so each also has a starred variant whose LLM judge decides whether to continue after a satisfactory object is found. The No-intent planner stops at the first such object, and all methods share one detector for explicit requests.

E. Human Interaction

We surveyed 10 participants on how much additional searching they would tolerate after an acceptable but sub-optimal object was found. Their responses correspond to approximately 1.5 additional rooms, or $T_h = 56$ s. The survey collected no personally identifying information and posed minimal risk. Participation was voluntary and informed. Human interaction is disabled in simulation. On hardware, the robot queries the user when the expected additional search time exceeds T_h.

V. Results

A. Simulation Results

1) AI Habitat: The oracle-perception results in Table III isolate grounding and planning from perception errors.

GRABS remains competitive on explicit requests. On HM3D, GRABS achieves 96.5% SR and 80.0% SPL, within 0.4% and 0.3% of the highest results, respectively. On MP3D, it ties the highest SR at 97.7% and achieves the highest SPL at 88.3%. These results show that graded candidates and belief-guided planning do not compromise standard object goal navigation.

Graded intent grounding provides substantial gains for vague-intent requests. Compared with the No-intent planner, GRABS improves Best accuracy by 27.8% absolute on HM3D and 29.6% on MP3D, while increasing mean satisfaction from 0.619 to 0.803 and from 0.588 to 0.880, respectively. The No-intent planner performs competitively on explicit requests but terminates after finding any satisfactory object, so it cannot distinguish an acceptable candidate from the best one for a vague request. GRABS also improves Best accuracy over the Landmark planner by 11.5% absolute on HM3D and 8.0% on MP3D. A single landmark cannot represent several categories that satisfy the intent to different degrees.

The time-aware MDP improves the quality–time trade-off. Compared with the Simple planner, GRABS achieves higher Best, Top-2, and mean satisfaction scores while reducing completion time from 140.3 to 98.4 s on HM3D and from 122.7 to 86.0 s on MP3D, a reduction of 29.9% in both cases. Accounting for future search outcomes and travel time thus yields more efficient room sequences than a heuristic score. The No-intent and Direct GPT planners terminate faster, but their lower Best and mean satisfaction scores show that this speed primarily results from accepting lower-quality candidates sooner rather than from a more effective search policy.

2) Isaac Sim: Isaac Sim adds the detection and verification uncertainty absent from the oracle setting, with results in Table IV.

VLM perception reduces absolute accuracy. Explicit

Fig. 4: Hardware navigation plan and execution for "I am very thirsty." The path is shown on the Isaac Sim digital-twin floor plan of the physical office. The robot finds no relevant item in the Break Room or Conference Room, then observes tea at A, coffee at B, and water at C.

SR decreases from 96.5% on HM3D and 97.7% on MP3D to 93.8% in Isaac Sim, while vague-intent Best accuracy decreases from 76.4% and 73.9% to 64.6%. Because environments and requests differ, this comparison indicates a trend rather than a controlled ablation.

The relative performance trend remains consistent under VLM perception. As in AI Habitat, GRABS achieves the highest explicit-request SR and SPL and the highest or tied-highest vague-intent metrics. It improves vague-intent Best accuracy by 27.1% absolute over the No-intent planner. Compared with the Simple planner, it maintains the same Best accuracy while increasing Top-2 accuracy from 81.2% to 83.3% and reducing completion time from 166.0 to 120.8 s. The benefits of graded intent grounding and belief-guided planning therefore persist under visual perception.

B. Hardware Results

Hardware behavior remains consistent with simulation. In all four vague-intent trials, GRABS retrieves the best available object, with a mean completion time of 182.8 s. Graded candidates let the robot treat the first acceptable object as a fallback rather than a stopping point, and belief updates and time-aware replanning transfer unchanged to real perception and navigation. For the explicit relational request, GRABS achieves an SPL of 32.8%, travels 71.8 m, and completes the task in 263.5 s. The request gives no room cue, so the robot searched other rooms before the belief update directed it to the office area, which explains the lower SPL. There it selects the silver trash bin near a door while rejecting a silver trash bin away from a door and a black trash bin, so VLM verification checks category, attribute, and spatial relation jointly on real images.

Human interaction supports time-aware search continuation. Figure 4 shows the complete execution for "I am very thirsty." The robot begins at the reception area and detects tea at observation point A. Tea satisfies the request but may not be the best available option. Because the expected time to find a better candidate exceeds T_h, the robot asks the user whether to continue. After confirmation, it resumes and detects coffee in the kitchen at point B. The remaining expected search time is then below T_h, so the robot continues without another query. It finally detects water in the dining area at point C, identifies it as the highest-satisfaction candidate, and terminates the search.

VI. Conclusion

We introduced GRABS, a belief-guided framework for best-available demand-driven navigation, which extends DDN to graded satisfaction and requires retrieving the highest-satisfaction object available. The robot continues searching while a better object is likely to be found within the user's patience, and asks the user otherwise. Experiments in AI Habitat, Isaac Sim, and a physical office confirm three properties. On explicit requests, GRABS matches standard object goal navigation performance. On vague requests, it retrieves the best available object more often than methods that stop at the first acceptable one. Its time-aware planning also reaches these objects faster than a heuristic planner over the same beliefs.

Several limitations remain. GRABS assumes a floor plan with semantically labeled rooms and reasons at room granularity, so it does not address search within a room. Sat-

isfaction probabilities and the patience threshold are fixed and do not adapt to individual users. Future work includes learning satisfaction values and a patience model adapted per user from interaction feedback, and combining room-level beliefs with frontier-based exploration when no floor plan is available.

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